

Skill 2

```
nishanth@LAPTOP-6Q3ON07H:~/OSLab$ nano skill2.q2.c
nishanth@LAPTOP-6Q3ON07H:~/OSLab$ gcc skill2.q2.c -o skill2.q2
nishanth@LAPTOP-6Q3ON07H:~/OSLab$ ./skill2.q2

Enter command: ls
Unknown command.

Enter command: help
Commands: help, start, stop, exit

Enter command: ls
Unknown command.

Enter command: ^C
nishanth@LAPTOP-6Q3ON07H:~/OSLab$ nano skill2.q2.c
nishanth@LAPTOP-6Q3ON07H:~/OSLab$ gcc skill2.q2.c -o skill2.q2
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shell> help
Available commands:
ls
pwd
help
exit

shell> ls
LinuxLab      exp1  exp6  form.c  program02  program24  program33  q3
MYProject    exp1.c exp6.c p1      program02.c program24.c program33.c q3.c
MyProject    exp2  exp7  p1.c   program1   program25  program4   q4
copy.c       exp2.c exp7.c p2     program1.c program25.c program4.c q4.c
```

```
shell> help
Available commands:
ls
pwd
help
exit

shell> hello
Command not found: hello

shell> exit
Shell terminated.
```

```
#include <stdio.h>
#include <string.h>
#include <unistd.h>
#include <termios.h>
#include <stdlib.h>

#define SIZE 100

int main()
{
    char input[SIZE];
    char ch;
    int i;
    struct termios old, new;

    tcgetattr(STDIN_FILENO, &old);

    new = old;
    new.c_lflag &= ~(ICANON | ECHO);

    tcsetattr(STDIN_FILENO, TCSANOW, &new);

    while (1)
    {
        i = 0;

        printf("\nshell> ");
        fflush(stdout);

        while (1)
```

```
{
    read(STDIN_FILENO, &ch, 1);

    if (ch == '\n' || ch == '\r')
    {
        input[i] = '\0';
        printf("\n");
        break;
    }

    if (ch == 127 || ch == 8)
    {
        if (i > 0)
        {
            i--;
            printf("\b \b");
            fflush(stdout);
        }
    }
    else
    {
        if (i < SIZE - 1)
        {
            input[i] = ch;
            i++;

            printf("%c", ch);
            fflush(stdout);
        }
    }
}
```

```
}

if (strcmp(input, "exit") == 0)
{
    break;
}
else if (strcmp(input, "help") == 0)
{
    printf("Available commands:\n");
    printf("ls\n");
    printf("pwd\n");
    printf("help\n");
    printf("exit\n");
}
else if (strcmp(input, "ls") == 0)
{
    system("ls");
}
else if (strcmp(input, "pwd") == 0)
{
    system("pwd");
}
else if (strlen(input) == 0)
{
    continue;
}
else
{
    printf("Command not found: %s\n", input);
}
}
```

```
tcsetattr(STDIN_FILENO, TCSANOW, &old);

printf("Shell terminated.\n");

return 0;
}
```